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6TH TH SEMESTER

END SEM EXAM

Roll No.....

[B.TECH. – IT]

(MAY- 2018)

IT-313 ARTIFICIAL ITELLIGENCE

Time: 3 Hours

Max. Marks: 70

Note: Attempt any 5 Questions. Each Question Carry Equal Marks. Assume suitable missing data, if any.

Q1

a) Define the term agent with the help of example? Discuss in detail at least about three types of agents? (2+6)

b) What are cuts in prolog? Explain the use of cuts in prolog step by step with the help of example? (2+4)

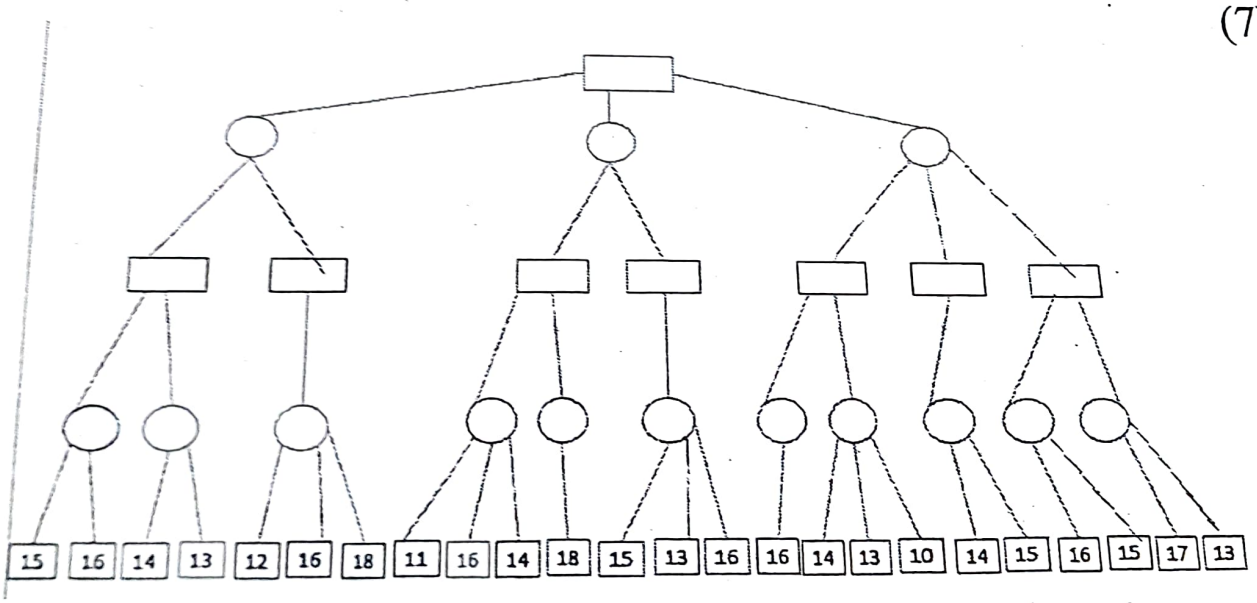
Q2

a) Show how the algorithm AlphaBeta explores the given game tree, searching from left to right.

(i) Fill in the leaves that are inspected by AlphaBeta.

(ii) Show the cutoffs and label them with their type.

(iii) Mark the move that AlphaBeta will choose for MAX at the root. (7)



What is learning by induction? Explain the concept of version spaces in detail with the help of example? (2+5)

P.T.O

Q3.

a) Explain each of the following forms of knowledge representation with the help of example? (3X3 = 9)

(i) Semantic Nets

(ii) Frames

(iii) Scripts

b) Write down the Unification Algorithm? Explain its use with the help of example? (2+3)

Q4. Translate the following sentence in to predicate logic and then covert them into clause form (7x2=14)

(i) All women's who like ice creams like chocolate.

(ii) No man is happy with a spendthrift wife.

(iii) The best movie in Hollywood is always better than the best movie in Bollywood.

(iv) Some people like eating outside all the time and some people like eating inside all the time.

(v) Feeling are just visitors. Let them come and go.

(vi) I will complete the work by Monday.

(vii) There are some dreams for which you can sacrifices anything.

Q5.

a) Explain each of the following with the help of example? (2X4 = 8)

(i) Forward Reasoning

(ii) Backward Reasoning

a) Discuss resolution procedure with the help of example? (

Q6.

a) Write a detailed note on following (7X2=14

(i) Natural language processing

(ii) Expert system